1. recall that dry air consists of gases, some of which are elements (for example oxygen, nitrogen and argon) and some compounds (for example carbon dioxide);
2. recall the symbols for the atoms and molecules of these gases in the air;
3. recall that most non-metal elements and most compounds between non-metal elements are molecular;
4. understand that some molecular elements and compounds have low melting and boiling points;
5. interpret qualitative and quantitative data about the properties of molecular elements and compounds, for example melting and boiling points;
6. understand that the elements and compounds in the air are gases because they consist of small molecules with weak forces of attraction between the molecules;
7. understand that pure molecular compounds do not conduct electricity because their molecules are not charged;
8. **understand that bonding within molecules is covalent and arises from the electrostatic attraction between the nuclei of the atoms and the electrons shared between them: covalent bonds are strong;**
9. translate between representations of molecules including molecular formulae, 2-D diagrams in which covalent bonds are represented by lines and 3-D diagrams for:
    - elements that are gases at 20°C;
    - simple molecular compounds;
10. recall that the Earth's hydrosphere (oceans) consists mainly of water with some dissolved compounds;
11. recall that sea water in the hydrosphere is 'salty' because it contains dissolved ionic compounds called salts;
12. understand that solid ionic compounds form crystals because the ions are arranged in a regular way;
13. understand that ions in a crystal are held together by the attraction between opposite charges: this is ionic bonding;
14. understand how the physical properties of solid ionic compounds (melting point, boiling point, electrical conductivity) relate to their giant, three-dimensional structures;
15. describe what happens to the ions when an ionic crystal dissolves in water;
16. explain that ionic compounds conduct electricity when dissolved in water because the ions are charged and they are able to move around independently in the liquid;
17. **be able to work out the formulae for salts in the sea given a table of charges on ions (for example sodium chloride, magnesium chloride, magnesium sulfate, potassium chloride and potassium bromide.).**